Practical 12: Time Series Analysis and Forecasting## Practical Information| Attribute | Details ||----------|---------|| **Practical Number** | 12 || **Title** | Time Series Analysis and Forecasting || **Duration** | 2 Hours || **Course Outcomes (COs)** | CO2, CO4, CO5 || **Description** | Analyze time series data and implement forecasting models. |## Key Topics Covered- Decomposition- ARIMA- SARIMA- Exponential smoothing- LSTM- Prophet---## Instructions for Students1. **Complete all tasks** outlined in this practical2. **Write your code** in the cells provided below3. **Add comments** to explain your code4. **Include proper output** (prints, visualizations, etc.)5. **Submit your completed notebook** before the deadline---## Your Work Starts HereUse the cell(s) below to write your implementation:

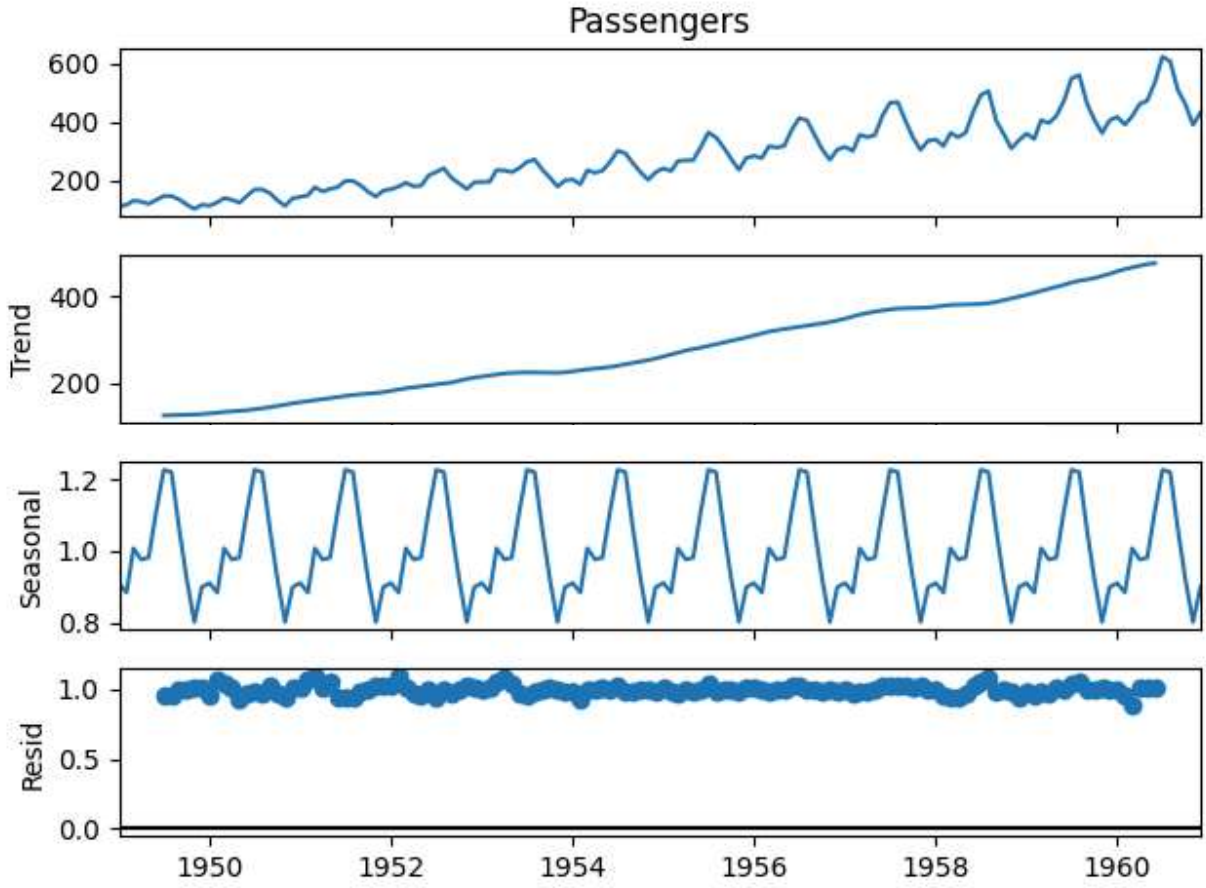
```
# Practical 12 - Code 1: Decomposition

import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset
url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/airline-pass
df = pd.read_csv(url, parse_dates=['Month'], index_col='Month')

# Decompose time series
decomposition = seasonal_decompose(df['Passengers'], model='multiplicative')
```

```
# Plot components
decomposition.plot()
plt.show()
```




```
# Practical 12 - Code 2: ARIMA Model

from statsmodels.tsa.arima.model import ARIMA
import numpy as np

# Split data
train = df[:'1958']
test = df['1959':]

# Build ARIMA model
model = ARIMA(train['Passengers'], order=(2,1,2))
model_fit = model.fit()

# Forecast
forecast = model_fit.forecast(steps=len(test))

# Plot
plt.figure(figsize=(8,5))
plt.plot(train.index, train['Passengers'], label='Train')
plt.plot(test.index, test['Passengers'], label='Test')
```

```
plt.plot(test.index, forecast, label='Forecast')
plt.legend()
plt.title("ARIMA Forecast")
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: V
  self._init_dates(dates, freq)
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: V
  self._init_dates(dates, freq)
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  self._init_dates(dates, freq)
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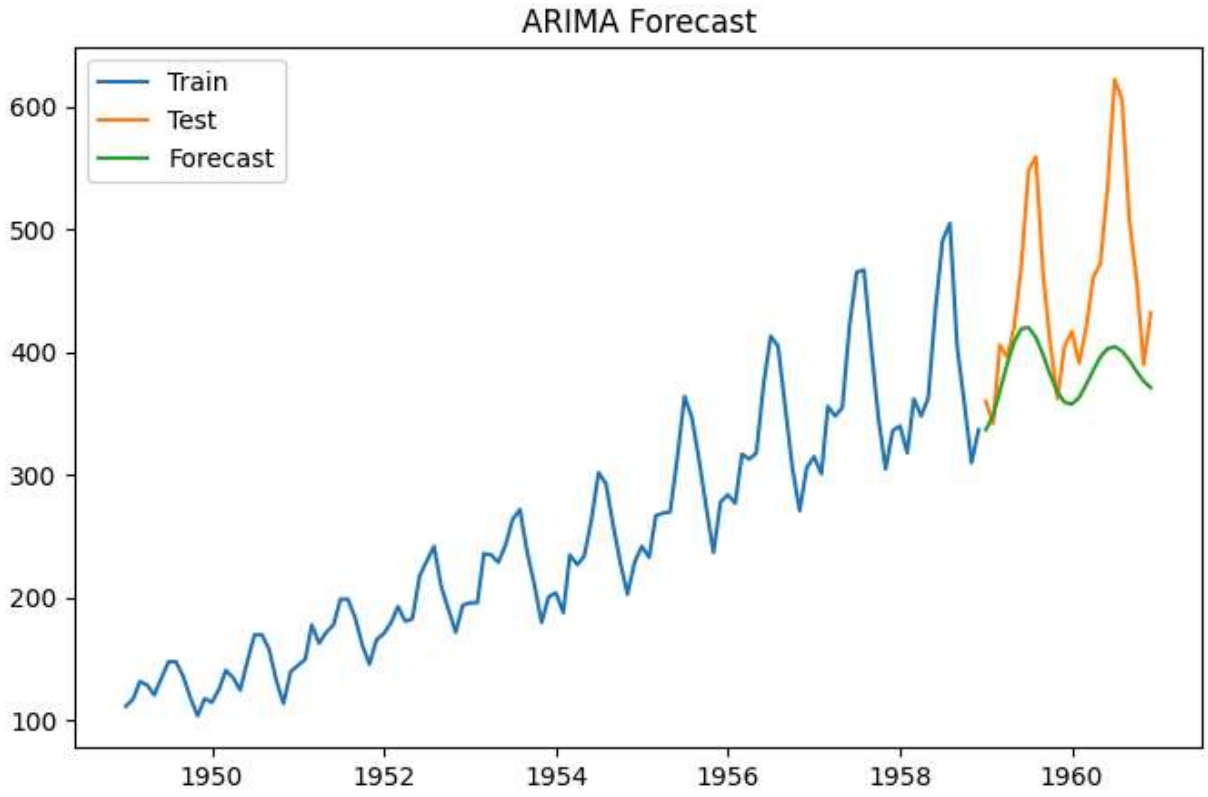
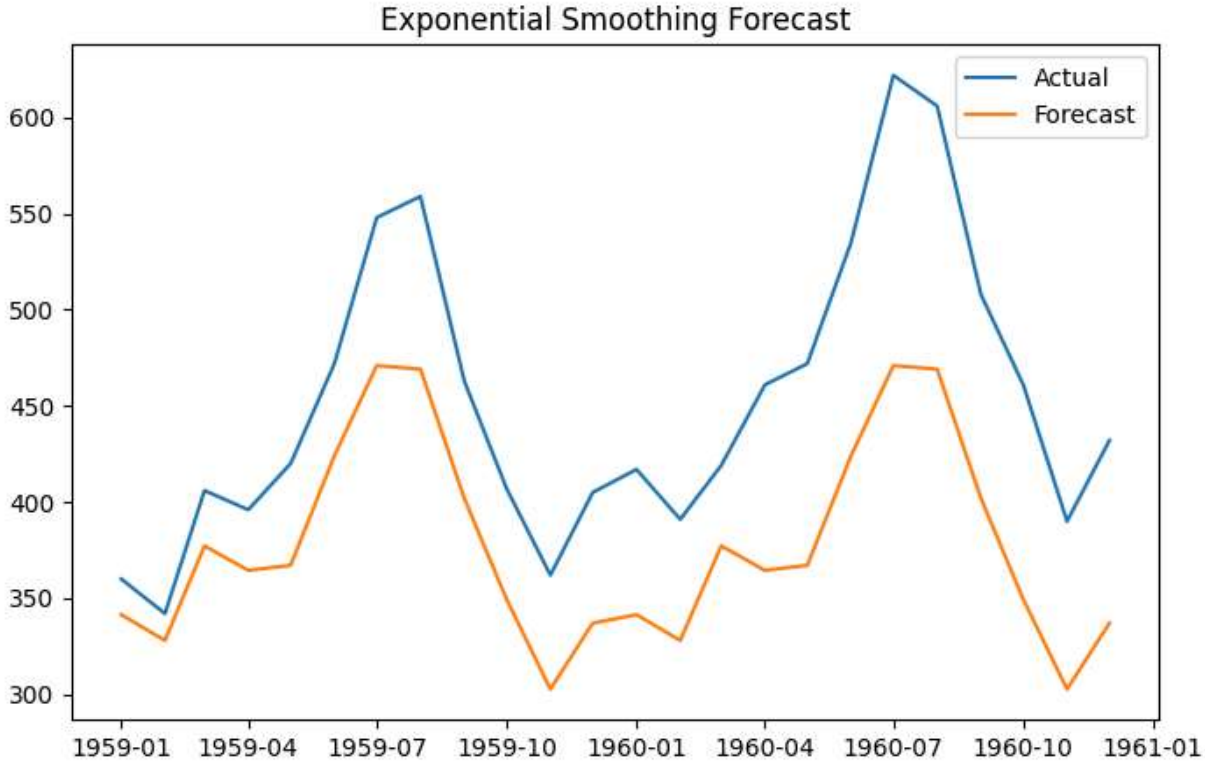



```
# Exponential Smoothing

from statsmodels.tsa.holtwinters import ExponentialSmoothing

model = ExponentialSmoothing(train['Passengers'], seasonal='mul', seasonal_peri
model_fit = model.fit()

forecast = model_fit.forecast(steps=len(test))

plt.figure(figsize=(8,5))
plt.plot(test.index, test['Passengers'], label='Actual')
plt.plot(test.index, forecast, label='Forecast')
plt.legend()
plt.title("Exponential Smoothing Forecast")
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: V
  self._init_dates(dates, freq)
```